Prepared For:

Project Assignment

# Brazilian E-Commerce Public Dataset by Olist

Data Analysis with SQL

By Aloysius Rabata Bayu Chrissanto



# Let’s Start With my Profile



## Aloysius Rabata Bayu Chrissanto

**Founder @Jalaninkedaikopi**




**Jl. Terogong Raya no 12A**

### EDUCATION

**Universitas Padjajaran (Unpad)** | Bandung, Jawa Barat | 2016-2020

- Faculty of Economics and Business
- Management (GPA: 3.45)
- Concentration: Marketing

### WORKING EXPERIENCE

**PT. Jalan Kita Bersama (@jalaninkedaikopi)** | Owner & CEO (Self-employed) | Dec 2022 – Now

- Starting from a small booth, I successfully scaled it into three active stores in Jakarta
- This entrepreneurial journey sharpened my skills in business strategy, operations management, branding, and financial control

**PT. Endorfin Entertainment Indonesia** | Head of Data & Reporting | Feb 2024 – Aug 2025

- Data Processing & Analytics
- Leading data & reporting teams to provide reports & data analytics for 12 regions
- Support SPG & Event Activation for PT. BAT (Cigarettes Product : Lucky Strike & Dunhill)

**PT. FIFGROUP Member of Astra** | Marketing Officer Development Program | Okt 2021 – Nov 2022

- Ranked 1 from 9 trainee on Marketing Development Program Okt 2021 Batch
- Promoted to Pontianak Area Marketing Officer
- Growing Pontianak Marketing Achievement on May 2022 to Nov 2022 from 1.1 Billion IDR to 1.9 Billion IDR

**IDStar Cipta Teknologi** | Digital Marketing Staff | Mar 2021 – Sept 2021

- Content Creation | Digital Marketing Strategy | Website Development
- Execute all social media activation & website development on IG / Tiktok/ Linked.in



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# Problem Statement

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## Data E-Commerce Sebagai Sumber Identifikasi Pola Penjualan

Industri e-commerce mengalami pertumbuhan yang pesat seiring meningkatnya aktivitas belanja online. Di Indonesia, nilai transaksi e-commerce pada 2018 diproyeksikan mencapai Rp144,1 triliun, meningkat dari Rp104 triliun pada 2017.Pertumbuhan tersebut menghasilkan data transaksi dalam jumlah besar yang dapat dimanfaatkan untuk memahami performa marketplace, pola transaksi, produk, serta distribusi seller dan customer.

Bagaimana data transaksi e-commerce dapat digunakan untuk mengidentifikasi pola penjualan dan memberikan insight yang mendukung pengambilan keputusan bisnis?

**Sumber:** Katadata, *Tiga Celah Pengembangan E-Commerce di Indonesia*, 26 Oktober 2018.

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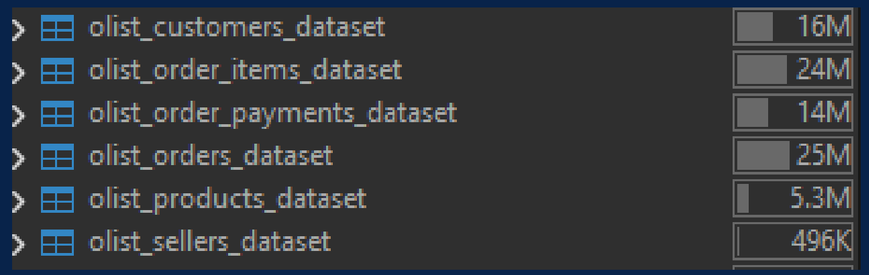


# Data Preparation

# Data Understanding

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**Data** yang digunakan merupakan Brazilian E-Commerce Public Dataset by Olist, yang didapatkan dari Kaggle.

Data dapat diunduh pada link berikut : Link Data Set

Data diolah menggunakan SQL dengan Postgree & DBeaver sebagai tools

| | order_id | customer_id | order_status | order_purc |
|---|---|---|---|---|
| 1 | e481f51cbdc54678b7cc49136f2d6a | 9ef432eb6251297304e76186b10a92 | delivered | 2017-10-02 10: |
| 2 | 53cdb2fc8bc7dce0b6741e21502734 | b0830fb4747a6c6d20dea0b8c802d7 | delivered | 2018-07-24 20: |
| 3 | 47770eb9100c2d0c44946d9cf07ec6 | 41ce2a54c0b03bf3443c3d931a3670 | delivered | 2018-08-08 08: |
| 4 | 949d5b44dbf5de918fe9c16f97b45f8 | f88197465ea7920adcdbec7375364d | delivered | 2017-11-18 19: |
| 5 | ad21c59c0840e6cb83a9ceb5573f81 | 8ab97904e6daea8866dbdbc4fb7aac | delivered | 2018-02-13 21: |
| 6 | a4591c265e18cb1dcee52889e2d8ac | 503740e9ca751ccdda7ba28e9ab8f6 | delivered | 2017-07-09 21: |
| 7 | 136cce7faa42fdb2cefd53fdc79a609 | ed0271e0b7da060a393796590e7b73 | invoiced | 2017-04-11 12: |

## Jumlah Data :

**Orders = 99,441**
Customers : 99,441
Order Items : 112,650
Payments : 103,886
Sellers : 3,095
Products : 32,951

| tanggal_awal | tanggal_akhir |
|---|---|
| 2016-09-04 21:15:19 | 2018-10-17 17:30:18 |

**Periode Data : 4 Sept 2016 sampai 17 Oktober 2018**

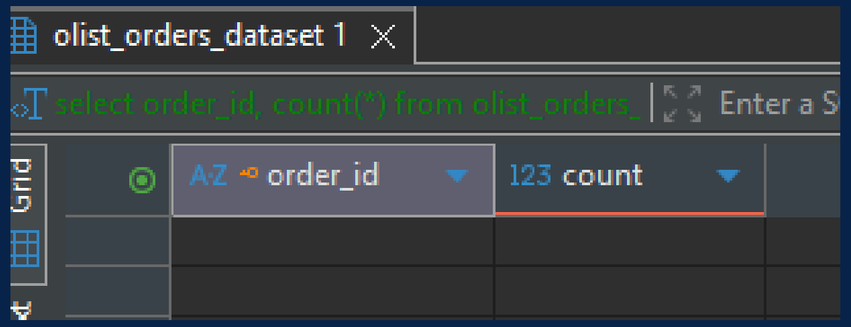
| Table | Column | Deskripsi |
|---|---|---|
| **Orders** | order_id | ID unik untuk setiap order/transaksi |
| | customer_id | ID customer yang melakukan order; digunakan untuk menghubungkan Orders dengan Customers |
| | order_status | Status order, seperti delivered, shipped, canceled, processing, dll. |
| | order_purchase_timestamp | Tanggal dan waktu ketika customer melakukan order |
| **Customers** | customer_id | ID customer yang digunakan untuk menghubungkan data customer dengan order |
| | customer_state | State/wilayah tempat customer berada |
| **Order Items** | order_id | ID order yang terkait dengan item yang dibeli |
| | product_id | ID produk yang dibeli |
| | seller_id | ID seller yang menjual produk |
| | price | Harga produk/item dalam transaksi |
| | freight_value | Biaya pengiriman untuk item |
| **Payments** | order_id | ID order yang terkait dengan pembayaran |
| | payment_type | Metode pembayaran, seperti credit card, boleto, voucher, atau debit card |
| | payment_value | Nilai pembayaran yang tercatat |
| **Products** | product_id | ID unik produk |
| | product_category_name | Kategori produk |
| **Sellers** | seller_id | ID unik seller |
| | seller_state | State/wilayah tempat seller berada |

# Check Duplicate & Missing

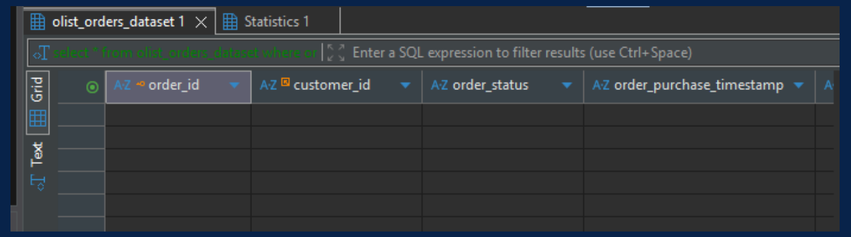
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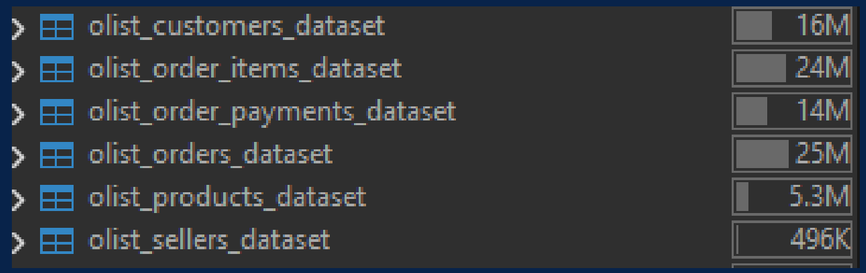
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```sql
-- check duplicate
select order_id, count(*)
from olist_orders_dataset
group by order_id
having count(*) > 1;
-- tidak memiliki duplicate
```




```sql
-- check missing
select *
from olist_orders_dataset
where order_id is null;
-- tidak memiliki missing value
```




Setelah dilakukan pengecheckan duplicate ditemukan duplicate pada :
order_id → payments_dataset
order_id → order_items_dataset

Dalam 1 order id bisa terjadi > 1 payment dan > 1 item

# Check Duplicate & Missing

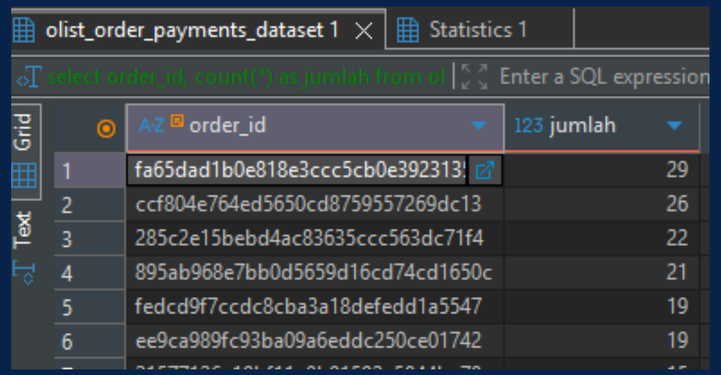
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```sql
-- Check duplicate order id di order payments
select order_id, count(*) as jumlah
from olist_order_payments_dataset
group by order_id
having count(*) > 1
order by jumlah desc;
-- Terdapat duplicate order_id, ada kemungkinan 1 order banyak payment
```

```sql
-- cek order id + payment_sequential untuk dijadikan composite PK
select
order_id,
payment_sequential,
count(*) as jumlah
from olist_order_payments_dataset
group by order_id, payment_sequential
having count(*) > 1;
-- tidak ada duplicate
```

```sql
-- check missing order id + payment squential
select *
from olist_order_payments_dataset
where order_id is null or payment_sequential is null;
-- tidak ada missing
```




Sebagai syarat penentuan PK, karena order_id tidak unique, kita menggunakan :

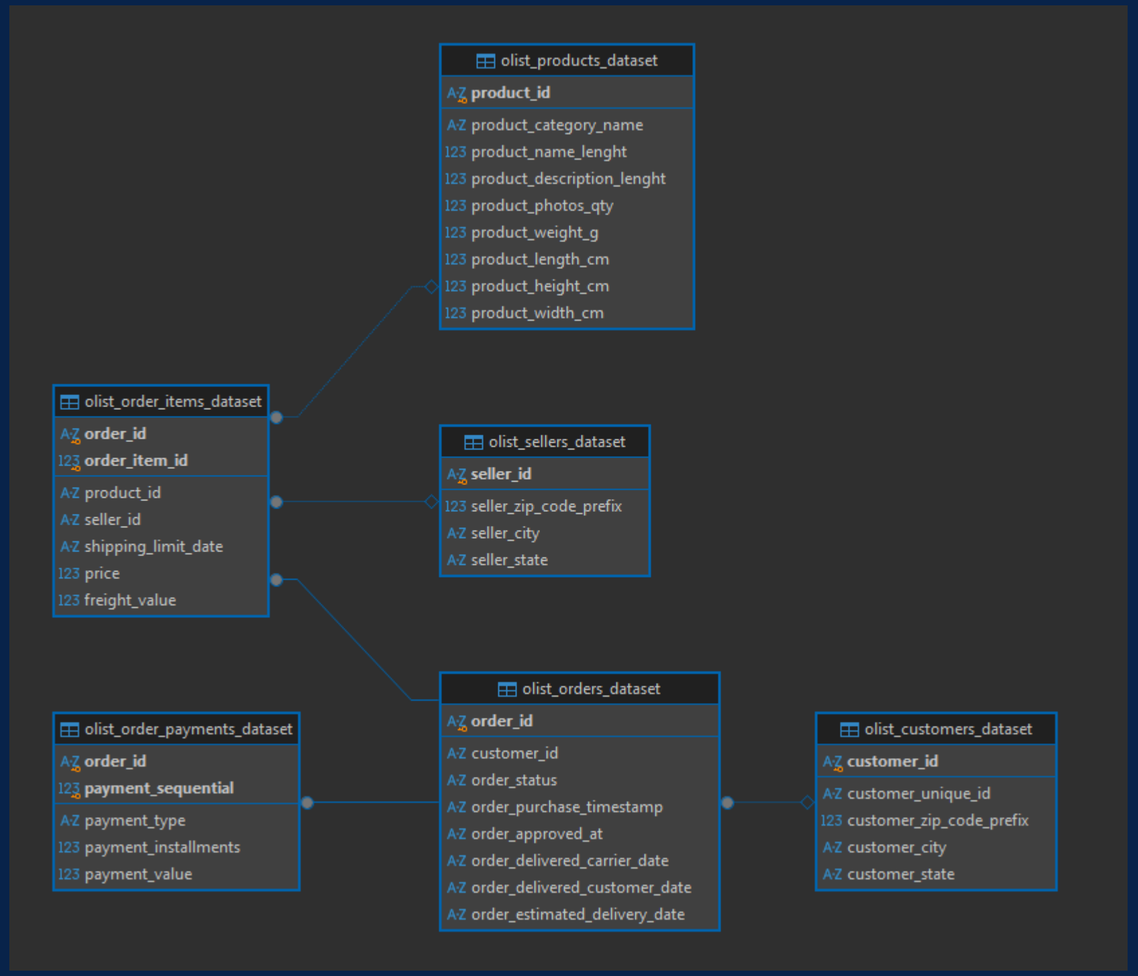
- order_id + payment_sequential pada order_payments_dataset
- Order_id + order_item_id pada order_items_dataset

Penggabungan 2 kolom, tersebut akan di set sebagai Composite PK pada tahap set PK dan FK

# Data Relationship

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Set Primary Key :

- Orders Dataset : Order_Id
- Order Item List : Order_id + Order_item_id
- Order Payments : Order_id + Payment_sequential
- Customers : Customer_id
- Sellers : Seller_Id
- Products : Product ID

Connect Foreign Key :

- Orders Dataset – Order Items Dataset → order_id
- Orders Dataset – Customers Dataset → customer_id
- Orders Dataset – Payments Dataset → order_id
- Orders Items Dataset – Sellers Dataset → seller_id
- Orders Items Dataset – Products Dataset → product_id

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# Data Exploration

# Order Status & Transaction type

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**Jumlah Data**
- Orders = 99,441
- Customers : 99,441
- Order Items : 112,650
- Payments : 103,886
- Sellers : 3,095
- Products : 32,951

**Periode Data**
**4 Sept 2016 sampai 17 Oktober 2018**

## Order by Status

| order_status | total_orders |
|---|---|
| delivered | 96,478 |
| shipped | 1,107 |
| canceled | 625 |
| unavailable | 609 |
| invoiced | 314 |
| processing | 301 |
| created | 5 |
| approved | 2 |

Berdasarkan table, mayoritas Orders memiliki status delivered dengan total order 96.478, sedangkan status lainnya memiliki jumlah yang jauh lebih kecil.

Dari data tersebut, didefinisikan bahwa order yang berhasil selesai adalah yang berstatus Delivered.

## Transaction Type

| payment_type | total_payments |
|---|---|
| credit_card | 76,795 |
| boleto | 19,784 |
| voucher | 5,775 |
| debit_card | 1,529 |
| not_defined | 3 |

Berdasarkan data, credit card merupakan metode pembayaran yang paling banyak digunakan, sedangkan metode lainnya jauh lebih sedikit digunakan.

Notes : ada 3 transaksi yang terdeteksi Not Defined

# Statistik Deskriptif

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## Rentang Price

olist_order_items_dataset 1

```
select min(price) as minimum_price, roun
```

Enter a SQL expression to filter results (use Ctrl+

Grid

| | 123 minimum_price | 123 average_price | 123 maximum_price |
|---|---|---|---|
| 1 | 0.85 | 120.65 | 6,735 |

## Rentang Freight

olist_order_items_dataset 1

```
select min(freight_value) as minimum_frei
```

Enter a SQL expression to filter results (use Ctrl+Spac

Grid

| | 123 minimum_freight | 123 average_freight | 123 maximum_freight |
|---|---|---|---|
| 1 | 0 | 19.99 | 409.68 |

Rentang Price cukup lebar dari 0.85 ke 6.735, dengan rata-rata 120,65.

Rentang Freight cukup lebar dari 0 – 409.68. dengan rata-rata 19.99.

olist_sellers_dataset 1

```
select seller_state, count(*) as total_sellers
```

Grid

Text

| | A-Z seller_state | 123 total_sellers |
|---|---|---|
| 1 | SP | 1,849 |
| 2 | PR | 349 |
| 3 | MG | 244 |
| 4 | SC | 190 |
| 5 | RJ | 171 |
| 6 | RS | 129 |
| 7 | GO | 40 |
| 8 | DF | 30 |
| 9 | ES | 23 |
| 10 | BA | 19 |
| 11 | CE | 13 |
| 12 | PE | 9 |
| 13 | PB | 6 |
| 14 | RN | 5 |
| 15 | MS | 5 |
| 16 | MT | 4 |
| 17 | SE | 2 |
| 18 | RO | 2 |
| 19 | MA | 1 |
| 20 | AC | 1 |
| 21 | PI | 1 |
| 22 | PA | 1 |
| 23 | AM | 1 |

Berdasarkan table Seller Paling banyak berasal dari SP, sebanyak 1.849.

Jumlah tersebut cukup mendominasi dibanding daerah lainnya.

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# Data Analysis

# Monthly Sales Performance

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olist_orders_dataset(+) 1

```
select date_trunc('month', o.order_purcha
```

Enter a SQL expression to filter result

Grid

Text

| | month | total_orders | total_sales |
|---|---|---|---|
| 1 | 2016-09-01 00:00:00.000 | 1 | 134.97 |
| 2 | 2016-10-01 00:00:00.000 | 265 | 40,325.1 |
| 3 | 2016-12-01 00:00:00.000 | 1 | 10.9 |
| 4 | 2017-01-01 00:00:00.000 | 750 | 111,798 |
| 5 | 2017-02-01 00:00:00.000 | 1,653 | 234,224 |
| 6 | 2017-03-01 00:00:00.000 | 2,546 | 359,204 |
| 7 | 2017-04-01 00:00:00.000 | 2,303 | 340,674 |
| 8 | 2017-05-01 00:00:00.000 | 3,546 | 489,348 |
| 9 | 2017-06-01 00:00:00.000 | 3,135 | 421,930 |
| 10 | 2017-07-01 00:00:00.000 | 3,872 | 481,615 |
| 11 | 2017-08-01 00:00:00.000 | 4,193 | 554,708 |
| 12 | 2017-09-01 00:00:00.000 | 4,150 | 607,405 |
| 13 | 2017-10-01 00:00:00.000 | 4,478 | 648,250 |
| 14 | 2017-11-01 00:00:00.000 | 7,289 | 987,749 |
| 15 | 2017-12-01 00:00:00.000 | 5,513 | 726,035 |
| 16 | 2018-01-01 00:00:00.000 | 7,069 | 924,631 |
| 17 | 2018-02-01 00:00:00.000 | 6,555 | 826,428 |
| 18 | 2018-03-01 00:00:00.000 | 7,003 | 953,340 |
| 19 | 2018-04-01 00:00:00.000 | 6,798 | 973,520 |
| 20 | 2018-05-01 00:00:00.000 | 6,749 | 977,530 |
| 21 | 2018-06-01 00:00:00.000 | 6,099 | 856,069 |
| 22 | 2018-07-01 00:00:00.000 | 6,159 | 867,945 |
| 23 | 2018-08-01 00:00:00.000 | 6,351 | 838,568 |

Penjualan menunjukan pertumbuhan dari 2016 – 2018, pertumbuhan terjadi secara signifikan pada tahun 2017, bahkan mencapai order tertinggi dengan total 7.289 order.

Sedangkan pada tahun 2018, jumlah order relative lebih stabil.

# Payment Method Performance

Prepared For:

olist_order_payments_dataset 1

```
select payment_type, count(*) as total_pay
```

Enter a SQL expression to filter results (use Ctrl+Space)

Grid | Text

| | payment_type | total_payments | total_payment_value | average_payment_value |
|---|---|---|---|---|
| 1 | credit_card | 76,795 | 12,542,100 | 163.32 |
| 2 | boleto | 19,784 | 2,869,360 | 145.03 |
| 3 | voucher | 5,775 | 379,437 | 65.7 |
| 4 | debit_card | 1,529 | 217,990 | 142.57 |
| 5 | not_defined | 3 | 0 | 0 |

Credit card merupakan metode pembayaran paling dominan dengan total transaksi sebesar 76.795 dan dengan payment value sebesar 12.542.100. Creditcard juga memiliki average payment value tertinggi dibandingkan tipe pembayaran lainnya, sebesar 163.32 per transaksi

# Seller Performance by State

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olist_sellers_dataset 1

select s.seller_state, count(distinct s.seller_ | Enter a SQL expression to filter results (use Ctrl+Space)

Grid | Text

| | seller_state | total_sellers | total_orders | total_sales |
|---|---|---|---|---|
| 1 | SP | 1,769 | 68,641 | 8,511,620 |
| 2 | PR | 335 | 7,512 | 1,232,050 |
| 3 | MG | 236 | 7,735 | 977,835 |
| 4 | RJ | 163 | 4,227 | 820,609 |
| 5 | SC | 184 | 3,603 | 613,594 |
| 6 | RS | 125 | 1,962 | 373,416 |
| 7 | BA | 18 | 550 | 277,925 |
| 8 | DF | 30 | 808 | 94,840.1 |
| 9 | PE | 9 | 403 | 91,163.6 |
| 10 | GO | 39 | 451 | 64,806.5 |
| 11 | ES | 22 | 310 | 46,941.6 |
| 12 | MA | 1 | 389 | 36,098 |
| 13 | CE | 12 | 87 | 19,517.8 |
| 14 | MT | 4 | 136 | 17,007.8 |
| 15 | PB | 6 | 35 | 16,395.5 |
| 16 | RN | 5 | 51 | 9,992.6 |
| 17 | MS | 5 | 49 | 8,551.69 |
| 18 | RO | 2 | 14 | 4,762.2 |
| 19 | PI | 1 | 11 | 2,383 |
| 20 | SE | 2 | 9 | 1,606.2 |
| 21 | PA | 1 | 8 | 1,238 |
| 22 | AM | 1 | 3 | 1,177 |

-- SP memiliki jumlah seller, jumlah order, dan total sales terbesar.
-- Seller dari SP menghasilkan 68,641 delivered orders
-- dengan total sales sebesar 8,511,620.
-- PR dan MG menjadi state berikutnya dengan total sales terbesar.

# Product Category Performance

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olist_products_dataset 1

```
select p.product_category_name, count(d
```

Enter a SQL expression to filter results (use Ctrl+Space)

Grid | Text

| | product_category_name | total_orders | total_items | total_sales |
|---|---|---|---|---|
| 1 | beleza_saude | 8,647 | 9,465 | 1,233,110 |
| 2 | relogios_presentes | 5,495 | 5,859 | 1,166,160 |
| 3 | cama_mesa_banho | 9,272 | 10,953 | 1,023,390 |
| 4 | esporte_lazer | 7,530 | 8,431 | 954,835 |
| 5 | informatica_acessorios | 6,530 | 7,644 | 888,710 |
| 6 | moveis_decoracao | 6,307 | 8,160 | 711,921 |
| 7 | utilidades_domesticas | 5,743 | 6,795 | 615,634 |
| 8 | cool_stuff | 3,559 | 3,718 | 610,211 |
| 9 | automotivo | 3,810 | 4,140 | 578,973 |
| 10 | brinquedos | 3,804 | 4,030 | 471,296 |
| 11 | ferramentas_jardim | 3,448 | 4,268 | 470,506 |
| 12 | bebes | 2,809 | 2,982 | 400,428 |
| 13 | perfumaria | 3,086 | 3,340 | 390,151 |
| 14 | telefonia | 4,093 | 4,430 | 309,861 |
| 15 | moveis_escritorio | 1,254 | 1,668 | 268,153 |
| 16 | papelaria | 2,264 | 2,466 | 223,790 |
| 17 | pcs | 177 | 199 | 218,684 |
| 18 | pet_shop | 1,688 | 1,924 | 211,696 |
| 19 | instrumentos_musicais | 611 | 651 | 184,316 |
| 20 | eletroportateis | 609 | 658 | 182,754 |

-- Insight:
-- Kategori beleza_saude menghasilkan total sales terbesar
-- sebesar 1,233,110 dari 9,465 items.
-- Kategori relogios_presentes berada di posisi kedua
-- dengan total sales sebesar 1,166,160.
-- Sementara itu, cama_mesa_banho memiliki jumlah item terjual
-- paling banyak di antara kategori teratas, yaitu 10,953 items.

# Customer Distribution by State

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| | customer_state | total_orders |
|---|---|---|
| 1 | SP | 40,501 |
| 2 | RJ | 12,350 |
| 3 | MG | 11,354 |
| 4 | RS | 5,345 |
| 5 | PR | 4,923 |
| 6 | SC | 3,546 |
| 7 | BA | 3,256 |
| 8 | DF | 2,080 |
| 9 | ES | 1,995 |
| 10 | GO | 1,957 |
| 11 | PE | 1,593 |
| 12 | CE | 1,279 |
| 13 | PA | 946 |
| 14 | MT | 886 |
| 15 | MA | 717 |
| 16 | MS | 701 |
| 17 | PB | 517 |
| 18 | PI | 476 |
| 19 | RN | 474 |
| 20 | AL | 397 |
| 21 | SE | 335 |
| 22 | TO | 274 |
| 23 | RO | 243 |
| 24 | AM | 145 |
| 25 | AC | 80 |
| 26 | AP | 67 |
| 27 | RR | 41 |

-- Insight:
-- SP memiliki jumlah delivered order terbesar yaitu 40,501 orders,
-- diikuti oleh RJ sebanyak 12,350 orders dan MG sebanyak 11,354 orders.
-- Hal ini menunjukkan bahwa sebagian besar order dalam dataset
-- berasal dari customer yang berlokasi di SP.

# Product Price Category

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| | price_category | total_items | average_price |
|---|---|---|---|
| 1 | Low Price | 39,024 | 31.3 |
| 2 | Medium Price | 51,171 | 92.87 |
| 3 | High Price | 22,455 | 339.25 |

-- Insight:
-- Sebagian besar item berada pada kategori Medium Price
-- dengan jumlah 51,171 items dan rata-rata harga 92.87.
-- Low Price memiliki 39,024 items, sedangkan
-- High Price memiliki jumlah paling sedikit yaitu 22,455 items.

# Thank You!

I KNOW THIS PRESENTATION IS FAR FROM PERFECT, BUT WITH DIBIMBING DATA ANALYST BOOTCAMP, I AM CONFIDENT TO MAKE A CLOSE TO PERFECT ANALYSIS AND STRATEGY IN THE FUTURE.

Aloysius Rabata Bayu Chrissanto

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